

Internship Report : Enhancing Extreme Rainfall Nowcasting with High-Resolution Lightning Data and Neural Networks

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This document contains the internship report under the subject Enhancing Extreme Rainfall Nowcasting with High-Resolution Lightning Data and Neural Networks, under the supervision of post-doc **Roberta BAGGIO**, baggio_r@univ-corse.fr, in Università di Corsica Pasquale Paoli.

1 Introduction and related work

Extreme weather events, such as intense precipitation, exert significant societal and economic pressures, rendering their accurate prediction a critical scientific challenge. While data-driven approaches—particularly neural networks—have demonstrated promising results in short-term forecasting (nowcasting up to 6 hours), the accurate prediction of localized extreme rainfall, especially over complex terrain, remains a formidable task.

This internship investigates the potential of high-resolution spatiotemporal electrical activity data (e.g., lightning) to enhance the forecasting of extreme rainfall events, with a specific focus on targeted sites in Corsica. When forecasting rainfall in the Mediterranean, selecting the appropriate neural network architecture to combine diverse data sources is crucial. Studies, like (Pujol et al., 2026) demonstrate that a hybrid modeling approach—integrating both direct observations and data from Numerical Weather Prediction (NWP) models—yields the most accurate results, particularly for predicting the most intense weather events. This hybrid methodology is supported by recent advances utilizing neural networks to combine NWP and observational data for both heavy Mediterranean rainfall prediction (Pujol et al., 2026) and local wind speeds at short time horizons (Baggio et al., 2025).

2 Methods employed

2.1 Building and organizing the data

2.1.1 The SAPHIR files

A model was developed for weather forecasting of some atmospheric variable (e.g. precipitation) at a specific station (Baggio et al., 2025). This model uses sources of different nature as input data: direct measurements from weather stations, predictions of NWP model at global scale (like ARPEGE model), or predictions of NWP model at regional scale (like AROME model, tailored to fine-scale local weather phenomena).

It has been validated using the MeteoNet dataset (Météo-France, 2020), created by Météo-France, a French public dataset combining observations, radar, satellite, and NWP prediction model outputs (AROME/ARPEGE) over two regions of France for the years 2016 to 2018, released to support research. ARPEGE is a global model that captures large-scale atmospheric patterns, whereas AROME is a high-resolution regional model tailored to fine-scale local weather phenomena. During the period covered by our dataset (2016–2018), both models generated forecasts through physics-based numerical simulation, without relying on machine learning. Moreover the MeteoNet dataset contains 51 weather ground stations in Corsica recording data such as temperature, wind speed, precipitation, air pressure, and more.

Observations and NWP predictions corresponding to a given ground station are, in a first step, regrouped in a single NetCDF file, which allows for storing multi-dimensional arrays indexed by time and coordinates. Each file corresponds to one ground station for a given day, containing the observations made by the station and the prediction made by AROME and ARPEGE. We will refer to those files in the rest of the document as SAPHIR files. Hybrid input data associated with a given forecasting task are then built from SAPHIR files, combining ground station observations and physical predictions, by passing them through an AI model, that we will treat in this report.

The objective of this internship is (1) to add to existing MeteoNet-based SAPHIR files a new data source that measures the electrical activities in the sky over Corsica stations, and (2) to check if this source improves the predictions of rainfalls. We will address these two tasks in the next section of the document.

2.1.2 The SAETTA instrument

At the laboratory, UMR SPE CNRS – University of Corsica, high resolution electrical activity data are available. These data are provided by SAETTA (Suivi de l'Activité Électrique Tridimensionnelle Totale de l'Atmosphère), an instrument installed and operated by LAERO in Toulouse (Coquillat et al., 2019) measuring the electric activities (or events as we will refer to in the rest of this document) of the sky, on a 240 km × 240 km squared centered on Corsica. We dispose of the coverage of stations covering all Corsica, from 2016 to 2023. The data is stored in .dat (a csv file) files, one file corre-

sponding to one day. An electric event is characterized by the following attributes :

- **time** : the time (in seconds) of the day the event was detected
- **lat, long, alt(m)** : the position in space of the detected event
- χ^2 , **nb_stations_detected** : a reconstruction quality indicator
- **P(dbW)** : event power in decibels, logarithmically scaled to optimize data storage.

We store events in a pandas DataFrame, enabling fast, vectorized data selection. To minimize detection errors, we retain only points detected by at least 7 stations and with a χ^2 index ≤ 0.5 – the latter measuring the discrepancy between the individual station measurements, with lower values indicating stronger agreement across stations. We additionally restrict the altitude to below 15 km.

2.1.3 Defining new variables in the SAPHIR files

We describe now the method we use in order to update the saphir files with lightning events. We update only files dated from April onwards, as SAETTA data is unavailable prior to that month. For a given station, we

consider the lightning events located in 10 km-square centered around the station. We define two new variables corresponding respectively to the number of events and to the corresponding total power :

```
saetta_events (time_saetta, altitude_level)
float64 ...
saetta_power (time_saetta, altitude_level)
float64 ...
```

Let s be a station, d a day, t a time interval, and l an altitude level. Let $f_s^{(d)}$ be a SAPHIR file corresponding to station s on day d

- $f_s^{(d)}[\text{saetta_events}][t, l] = v$ indicates that v electric events occurred within a $10 \text{ km} \times 10 \text{ km}$ area centered on station s during time interval t at altitude level l .
- $f_s^{(d)}[\text{saetta_power}][t, l] = p$ denotes the cumulated power of these events in decibels (dBW) received during the same time interval t and altitude level l .

In line with station data, we consider 6 minute time intervals before each t value (running to 1 day in the past and 3 days in the future with respect to the considered day d), and categorize events using specific altitude levels as given in the table below:

2.1.4 Updating the SAPHIR files

With the data structure established, we must now define an efficient strategy for updating the SAPHIR files. Given the substantial data volume—14 GB for SAPHIR and 12 GB for SAETTA—we need a high-performance

Table 1: Altitude level mapping

Index	Altitude Range (m)
0	0 – 3,000
1	3,000 – 6,000
2	6,000 – 9,000
3	9,000 – 15,000

approach to ensure rapid processing. **First step** : We need to initialize the saetta variables inside the SAPHIR files, so we assign new variables for each saphir file after April first, and initialize each value of the array to 0.

Second step : We extract station coordinates from the SAPHIR files and use the Python package *geopy* to define 10 km bounding boxes around each site. These bounds will enable us in the next step to map SAETTA events to their respective stations. By leveraging the vectorized capabilities of ‘pandas’, we perform these spatial lookups rapidly.

Third step: For each SAETTA file, we look at the locations of each events, and by using the bounding boxes we calculated before, we associate the stations to the SAETTA file. At the end of this step we have a dictionary with keys are the SAETTA files, and the values all the SAPHIR files we will need to update for this given SAETTA file.

We have chosen to update the saphir files per SAETTA file, because SAETTA file can contain more than 1 million events, opening one file can take time. The strategy to open a SAETTA file once is the best strategy.

When reading a SAETTA file, we create two new columns : one transforming the altitude in meters into the correct altitude index, and another on transforming the time in seconds, into a 6 minute time stamp.

Fourth step: We perform a nested loop. For each SAETTA file and for each associated SAPHIR file, we perform a filter to select the events taking place inside the SAPHIR station bounding box. Then we perform the aggregation and update. We group with a groupBy the extracted data by time interval and altitude level, sum the event counts, and compute the aggregate power (in dBW) using a *logsumexp* method. These values are then added to the *saetta_events* and *saetta_power* variables in the targeted SAPHIR file.

To aggregate the log-scaled power values, we use the *logsumexp* approach, which avoids numerical underflow or overflow when summing power values expressed on a logarithmic (dB) scale.

Those steps were executed on the SPE laboratory cluster, allowing programs to run in background, using parallelism to compute fast and efficiently the updates.

2.2 Final step : building data dictionaries

After updating the SAPHIR files, there remains one last step to use the data to train a model. This step consists of building dictionaries of data that directly contain, for each station, the input and output matrices (or “tensors”) of a given prediction task. They are stored in .h5 files.

Let d be a data source, s a station, and h an hour of the day. The dictionary corresponding to the data source d possesses a key s_d associated with a tensor. This tensor contains variables corresponding to the data source, aggregated over one hour, for the corresponding station plus 10 of its neighbors. These variables are listed in Table 2. The quantity we ultimately want to predict is the hourly cumulative precipitation at the target station.

Table 2: Variables and tensor shape associated with each data source.

Data source	Variables	Shape
Station	temperature	(7, 33)
	humidity	
	wind_speed	
	wind_direction	
	precipitation	
AROME	relative_humidity	(6, 7, 11, 11)
	wind_speed	
	wind_direction	
	temperature_2m	
	sealevel_pressure	
	dewpoint_2m	
	total_precipitation	
ARPEGE	wind_speed_	(6, 8, 7, 5, 5)
	wind_direction	
	vertical_velocity	
	temperature	
	pressure	
SAETTA	saetta_events	(7, 11, 2, 4)
	saetta_power	

The SAETTA data source is the one we added, its data structure (7, 11, 2, 4) captures:

- 7 forecast horizons
- 11 stations (local + 10 neighbors)
- 2 variables (events count, power in dBW)
- 4 altitude levels (0–3, 3–6, 6–9, 9–15 km)

Now, we finally possess data that we can use to train nowcasting models. The task is framed as a supervised learning problem, in which the loss (defined below) is minimized on a training subset (80% of the samples) and evaluated on the remaining validation and test subsets (20% of the samples) in order to monitor and avoid overfitting. From now on, we will focus entirely on how the models are implemented, and discuss their performances. The data sources are passed through a neural network using a probabilistic model, that we define in the next section.

3 Probabilistic Forecasting Model

The model’s goal is to provide nowcasting for rainfalls. In 2010, the WMO Working Group on Nowcasting Research defined nowcasting as “forecasting with local detail, by any method, over a period from the present to six hours ahead, including a detailed description of the present weather”¹. Given a time t in hours, the model provides nowcasting for 6 time horizons, from $t + 1$ to $t + 6$ hours.

Prior to model input, all feature tensors are standardized component-wise using mean and variance computed from the training subset to ensure numerical stability and balanced gradient flow during training.

3.1 Probabilistic Approach

As rainfall are rare events, most of the cumulative rainfalls is 0. Rainfall is often modeled using a mixed conditional distribution, combining a discrete component with a continuous one. Let’s denote by Y_{t+h} the random variable counting the 1 hour cumulative rainfall upcoming at horizon h . The discrete part of the distribution models the probability of having rain or not, and the continuous part models the cumulative rainfall amount when it rains.

Our goal is to produce a probabilistic forecast, i.e. to estimate the conditional probability density of Y_{t+h} , conditionally on the observation of the input tensor X_t , which represents all the information available at time t (station observations, AROME/ARPEGE fields, SAETTA lightning data). The predicted distribution is therefore conditional on X_t . Let $\mathbb{P}$ denote the probability measure.

We consider mixed conditional distributions of the general form:

$$\mathbb{P}(Y_{t+h} = 0) = 1 - p_{\text{wet}}, \quad \mathbb{P}(Y_{t+h} > 0) = p_{\text{wet}}$$

When $Y_{t+h} > 0$, the rainfall intensity follows a continuous distribution with density $f_+(y; \Pi_t^+)$ that can be chosen to be a lognormal distribution, with $\Pi_t^+ = (\mu_t, \sigma_t)$:

$$f_+(y; \Pi_t^+) = \frac{1}{y\sigma_t\sqrt{2\pi}} \exp\left(-\frac{(\log y - \mu_t)^2}{2\sigma_t^2}\right)$$

The resulting mixture conditional distribution can be written as :

$$f(y, \Pi_t) = (1 - p_{\text{wet}})\delta_0(y) + p_{\text{wet}}f_+(y; \Pi_t^+)1_{\{y>0\}}$$

, where δ_0 denotes the Dirac mass at zero and $\Pi_t = (\hat{p}_{\text{wet},t}, \hat{\mu}_t, \hat{\sigma}_t)$.

The conditional distribution is characterized by 3 parameters : p_{wet} , μ and σ^2 . Training the model will

¹World Meteorological Organization, Working Group on Nowcasting Research, 2010.

mean trying to fit those parameters given observations, to estimate a distribution corresponding to each horizon.

The network M outputs :

$$M(X_t; \hat{\theta}) = \hat{\Pi}_t,$$

where X_t denotes the covariates (station observations, AROME/ARPEGE fields, SAETTA lightning data) available at time t , and $\hat{\theta}$ the learned network parameters.

The optimal parameters $\hat{\theta}$ are learned by minimizing the negative log-likelihood over the observed values, evaluated on a training set of N samples over H horizons:

$$\mathcal{L}_{LL} = -\frac{1}{N} \sum_{k=1}^N \sum_{h=1}^H \ln f(Y(t_k + h); \hat{\Pi}_{t_k}).$$

We will use this loss as a metric later to assess the quality of our probabilistic forecasting model; in this context, this loss is often called the “logScore”. Since we are also interested in predicting extreme events of rainfalls, we will consider high threshold exceedance events and use binary classification scores, as explained in the next section.

3.2 From an arbitrary quantile level q to a binary classification

The rarer an event, the higher the quantile level required to characterize it: a quantile q close to 1 corresponds to a threshold that is only exceeded by a small fraction $(1 - q)$ of the rainfall distribution, i.e. to rare, extreme rainfall events. Conversely, lower values of q correspond to more common rainfall amounts. This makes the quantile level q a natural way to parametrize the rarity of the event we wish to predict.

Building on this, we reformulate the evaluation of rare event prediction as a binary classification problem: for a given quantile level q , we ask whether the observed rainfall exceeds the q -th quantile of the predicted distribution. This turns the original regression-like task – predicting a full rainfall distribution – into a series of binary questions of varying difficulty, each one targeting a different level of rarity. As q increases, the associated classification task becomes harder, since it focuses on rarer and less frequent rainfall events, providing a natural way to assess how well the model performs specifically on extreme events.

3.2.1 Step 1 – From quantile to threshold

The level q defines a concrete threshold Q_q through the climatological CDF F_C of the target variable:

$$F_C(Q_q) = q \iff Q_q = F_C^{-1}(q),$$

estimated in practice as the empirical q -th quantile of the observed rainfall values over the reference climatology. A larger q yields a higher, rarer threshold Q_q ; a smaller q yields a lower, more frequent one.

3.2.2 Step 2 – Binary label from the observation

Each continuous observation $Y(t + h)$ is converted into a binary label with respect to Q_q :

$$I_{t+h}(Q_q) = \begin{cases} 1 & \text{if } Y(t + h) \geq Q_q \quad (\text{event}), \\ 0 & \text{otherwise} \quad (\text{no event}). \end{cases}$$

3.2.3 Step 3 – Binary label from the predicted distribution

The network does not predict I_{t+h} directly; it predicts the parameters $\hat{\Pi}_t = (\hat{p}_{wet}, \hat{\mu}, \hat{\sigma})$ of the full conditional distribution. From this distribution, the probability of exceeding the same threshold Q_q is obtained by integrating the predicted density over the upper tail:

$$\hat{p}_t(Q_q) = \int_{Q_q}^{\infty} f(y; \hat{\Pi}_t) dy$$

This continuous probability is then thresholded at an operating point p^* to obtain a predicted binary label:

$$\hat{I}_{t+h}(Q_q) = \begin{cases} 1 & \text{if } \hat{p}_t(Q_q) > 1 - q, \\ 0 & \text{otherwise.} \end{cases}$$

3.2.4 Step 4 – Binary verification scores

For a fixed quantile level q , the procedure above yields, for every sample (station, horizon), a pair of binary labels $(I_{t+h}(Q_q), \hat{I}_{t+h}(Q_q))$: one observed, one predicted from the fitted distribution. Comparing these pairs over the whole test set defines the classification problem at level q , from which binary verification metrics can be computed.

A first, very common metric is the F1-Score, the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}.$$

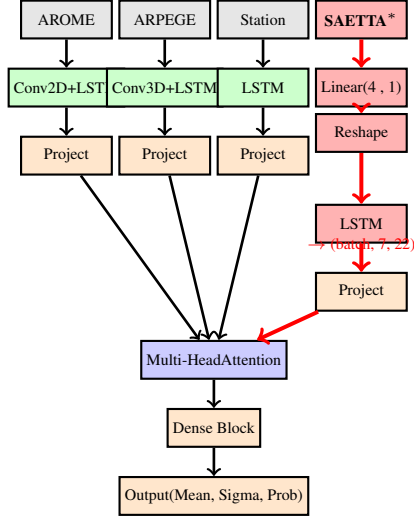
However, since intense rainfall events are rare compared to dry or low-rainfall periods, the class distribution of $I_{t+h}(Q_q)$ becomes strongly imbalanced as q increases, which limits the discriminative power of the F1-Score. We therefore also use the Peirce Skill Score (PSS), which measures the discrimination ability of the forecast independently of the class imbalance between rain and no-rain events:

$$\text{PSS} = \text{HR} - \text{FA} = \frac{\text{TP}}{\text{TP} + \text{FN}} - \frac{\text{FP}}{\text{TN} + \text{FP}},$$

where HR is the hit rate (probability of detection) and FA the false alarm rate. Contrary to the negative log-likelihood, we aim to maximize the PSS.

4 Neural Network Architecture

The data is fitted through a Neural Network, which we describe in the following image.



Multi-modal fusion with attention. * SAETTA: altitude dimension reduced from 4 to 1 via Li

All the data sources are independent and coming from different branches. This allows us to select freely which data sources we wish to use, which is useful in our case, as we aim to see if adding lightning to the sources improves the predictions, by comparing how models perform with or without this source.

4.1 Estimating the uncertainty

This model is highly sensible to the seed used to randomly initialize the weights. The results of a model could change drastically given different seeds. We need to evaluate the uncertainty between those models.

We define the following method: instead of training a model once, we train it 5 times, initializing its weights with 5 different seeds. The models are then saved one by one. Then, when computing the results by evaluating a metric, we aggregate by computing the mean and the standard deviation of those models, thus estimating the uncertainty. We will see the mean and the standard deviation on the plots showing the metrics.

5 Results

5.1 Models trained

We have trained two models, which we will present the results, by showing plots of various metrics.

1. **Baseline Corsica:** A model trained on meteorological data over Corsica, combining station observations and Météo-France sources (AROME/ARPEGE), but excluding lightning events. It is represented in **green** (—) on the plots.
2. **Baseline Corsica + SAETTA:** The model described above, augmented with the lightning events (SAETTA) source. It is represented in **blue** (—) on the plots.

5.2 Negative loglikelihood metric

The model uses a probabilistic approach to predict events, using the negative loglikelihood as a loss to minimize.

We address in this section the values of this loss for the models tested.

Negative Loglikelihood across models on test set

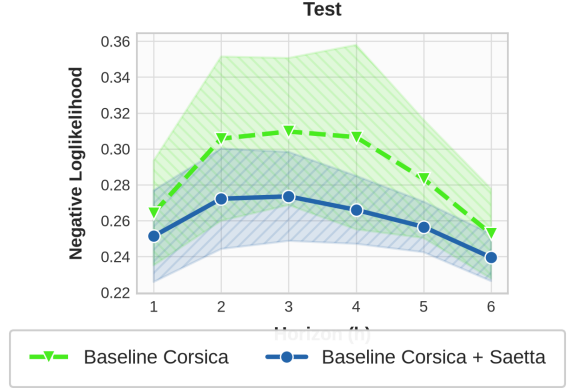


Figure 1: Negative loglikelihood

Figure 1 illustrates the hierarchy among the models. First, the model trained on Corsica without the SAETTA source exhibits a higher negative log-likelihood than the model trained with SAETTA, across all horizons. Furthermore, the baseline (without SAETTA) shows a wider confidence band than the SAETTA-augmented model, indicating that its results are less stable across seeds.

Second, the transfer learning model outperforms Corsica + SAETTA, achieving a lower mean negative log-likelihood. Its confidence band lies mostly below that of Corsica + SAETTA, with only slight overlap, confirming its superior and more consistent performance.

Based on Figure 1, we can conclude that, across all horizons and in terms of negative log-likelihood, Corsica + SAETTA has better performances than Corsica Baseline.

We will now focus on comparing the capacity of the models to predict rarer events. This task falls onto a classification problem, and we will use classification metrics, such as the F1-Score and the PSS.

5.3 F1-Score

Now, let's examine the F1-Score of the models.

As shown in Figure 2, the different models achieve nearly identical F1-scores across all horizons, indicating that this metric does not allow us to establish a clear hierarchy between the models. This lack of discriminative power can be explained by the strong class imbalance in the rainfall data: since intense rainfall events are rare compared to dry or low-rainfall periods, the F1-score is dominated by the majority (no-event) class and becomes largely insensitive to the differences in predictive skill that the PSS is able to capture. In other words, a model can achieve a high F1-score simply by correctly classifying the abundant negative examples, even if it

F1 Score across models on test set

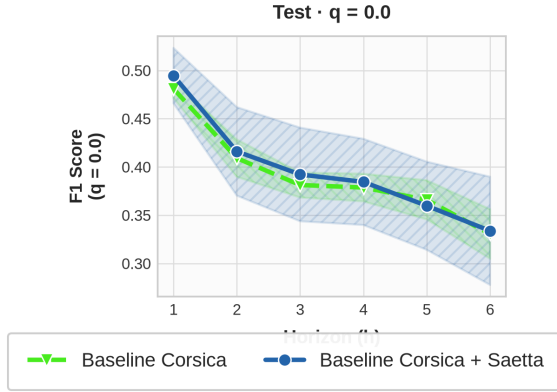


Figure 2: F1-Score across models

performs poorly on the rare positive (extreme rainfall) events that are of primary interest here. This metric did not allow us to really make a hierarchy between the models, in the next section we use another metric, the PSS (Pierce Skill Score).

5.4 Peirce Skill Score

The PSS was calculated onto 4 levels of quantiles $q = [0, 0.6, 0.8, 0.95]$, and plotted on Figure 3

If the measurements over the negative log-likelihood permitted us to establish the hierarchy :

$$\text{Corsica} + \text{SAETTA} > \text{Corsica without SAETTA},$$

looking at the PSS makes the observations more nuanced.

Quantile $q = 0$ (rain occurrence). For $q = 0$, which corresponds to answering the question “does it rain?”, the Corsica + SAETTA model fails to improve upon Corsica without SAETTA, the two models have very similar results.

Higher quantiles (rarer events). For $q \in \{0.6, 0.8, 0.95\}$, the PSS of Corsica + SAETTA improves while that of Corsica Baseline declines, widening the performance gap between the two models.

6 Conclusion

Exploring these three metrics allowed us to compare and rank the trained models, and to draw several conclusions from the results. For clarity, the plots presented here are restricted to the models trained over Corsica.

Regarding the binary question “will it be raining in x hours?”, adding a lightning-based data source does not improve performance: as shown earlier, the Corsica baseline and Corsica + SAETTA models achieve similar results on this task. Where SAETTA does make a difference is in predicting rarer rainfall events: the performance of Corsica + SAETTA improves noticeably as

PSS — Peirce Skill Score across models on test set

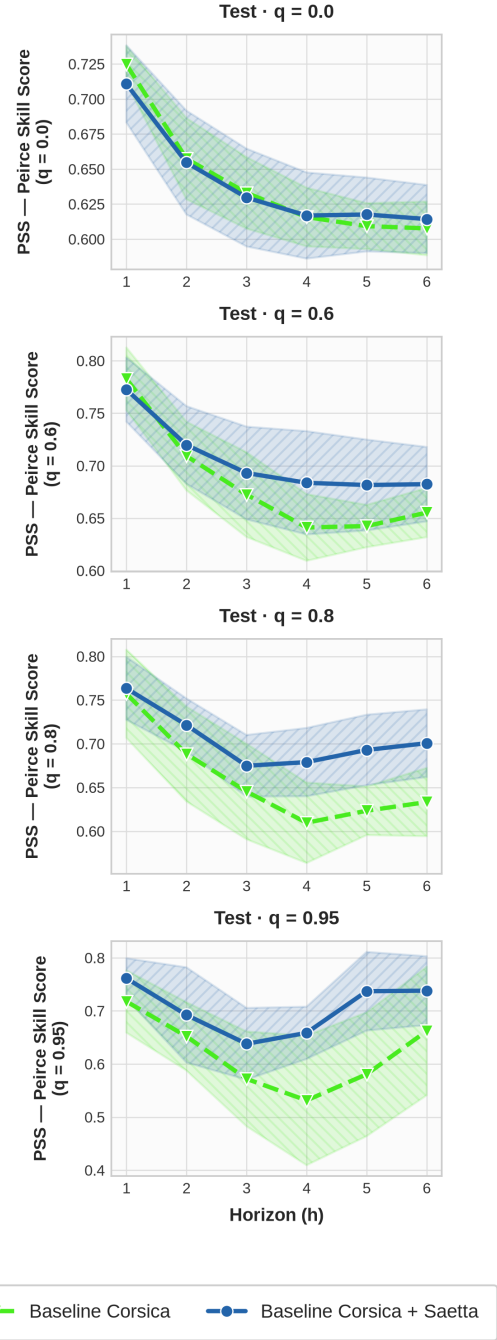


Figure 3: PSS over different models

the probability quantile increases, gradually approaching – and even matching – the Corsica baseline at the last horizon.

We also experimented with transfer learning, using a model pretrained on data covering the whole of South-Eastern France. Its weights were frozen and the model was subsequently trained on SAETTA data alone. Con-

trary to our expectations, this approach did not benefit from the larger and more diverse pretraining dataset: the model overfitted and ultimately performed worse than the South-Eastern France model trained without SAETTA. This suggests that freezing the pretrained weights was too restrictive to allow the model to properly integrate the new lightning-based modality, and that training only one data source is not sufficient to improve performances.

References

- B. Baggio et al. 2025. Local wind speed forecasting at short time horizons based on numerical weather prediction and observations from surrounding stations. *Journal of Geophysical Research*.
- Sylvain Coquillat, Eric Defer, Pierre de Guibert, Dominique Lambert, Jean-Pierre Pinty, Véronique Pont, Serge Prieur, Ronald J. Thomas, Paul R. Krehbiel, and William Rison. 2019. [SAETTA: high-resolution 3-D mapping of the total lightning activity in the Mediterranean Basin over Corsica, with a focus on a mesoscale convective system event](#). *Atmospheric Measurement Techniques*, 12(11):5765–5790.
- Météo-France. 2020. Meteonet, an open reference weather dataset by meteo france. <https://meteofrance.github.io/meteonet/>. Accessed: 2026-07-23.
- A. Pujol et al. 2026. Improving prediction of heavy rainfall in the mediterranean with neural networks using both observation and numerical weather prediction data. *Artificial Intelligence for the Earth Systems*.

Supplementary Figures This section contains figures to understand better this project

A Data Visualization and Architecture

The plot below illustrates the integration of SAETTA lightning data into SAPHIR files. We present here a SAPHIR file for August 6th, 2018, from station 20023005. Each subplot corresponds to one altitude level, with two variables displayed: the blue curve represents the number of lightning events detected during that day, while the orange curve shows the cumulative power emitted by those events. Unlike continuous meteorological variables such as temperature, wind speed, or rainfall, lightning events are inherently sparse—SAETTA variables are zero for most of the time interval. To improve readability, we only plot non-zero values and mark these detection periods with red vertical bars, thereby omitting intervals where no activity occurred.

A.1 Altitude Plots

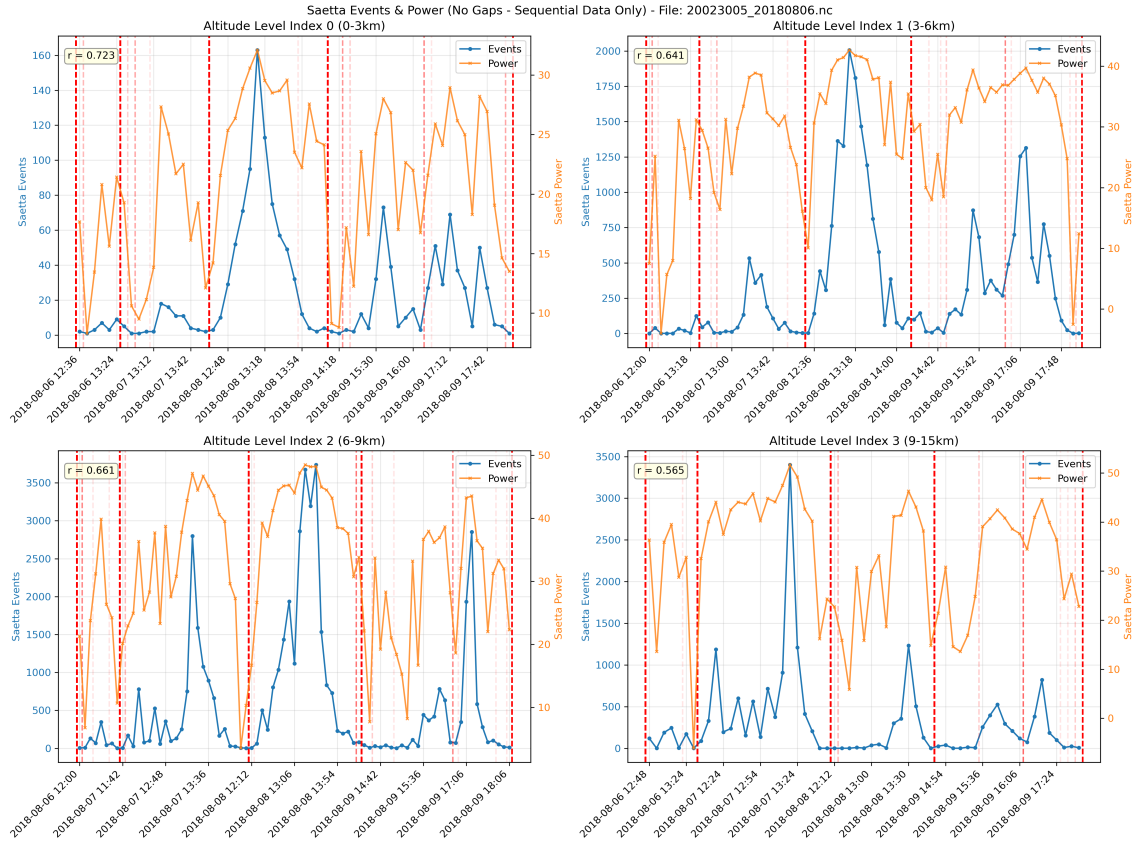


Figure 4: Lightning events on August 6th, 2018

A.2 SAETTA Lightning Events

The map below visualizes all recorded lightning events on August 7, 2018. Orange clusters indicate lightning strikes, while green squares represent the detecting stations and their coverage areas.

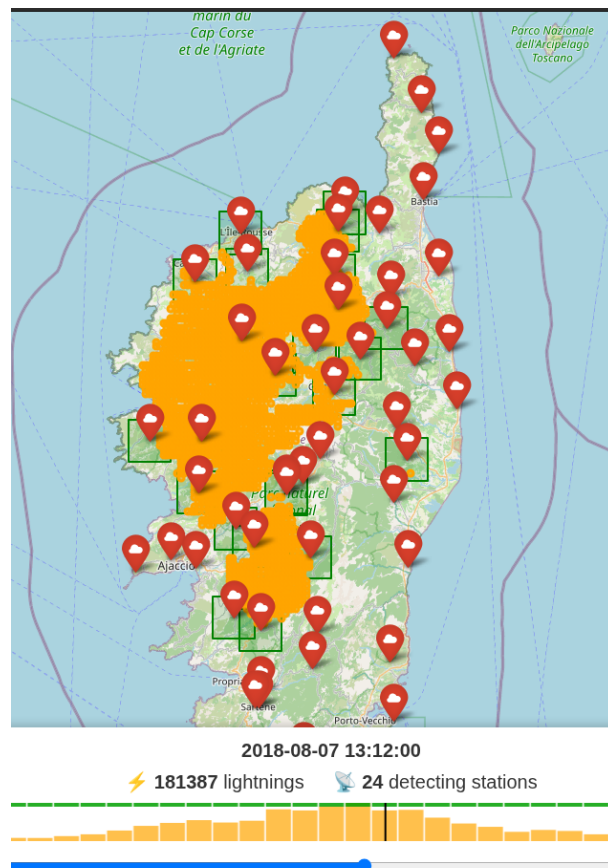


Figure 5: Raw SAETTA lightning detection with geographic reference points.